INJECTIVITY IS EQUIVALENT TO VANISHING OF FIRST EXT AGAINST EVERY MODULE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The long exact sequence of Ext groups associated with a short exact sequence. Algebra, Lemma \ref{algebra-lemma-reverse-long-exact-seq-ext}, cited at proof lines 13681-13682. Provides the exact sequence whose exactness at the Hom term is equivalent to injectivity.
- Identification of the Ext groups with the derived functors of Hom, and vanishing of the higher ones for injective modules. Homology, Lemma \ref{homology-lemma-characterize-injectives} and Lemma \ref{lemma-relation-ext-ext}, cited at proof lines 13686-13688. Give the converse implication.

Source reference. Original lemma 0AVE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13682–13682. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 065P: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7908–7908. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0136: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1424–1424. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0AUL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13654–13654. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let J be an R-module. The following are equivalent*

(1) *J is injective,*
(2) *$\mathrm{Ext}^1_R(M, J) = 0$ for every R-module M.*

Proof. Let $0 \to M'' \to M' \to M \to 0$ be a short exact sequence of R-modules. Consider the long exact sequence

$$\begin{aligned} 0 \to \mathrm{Hom}_R(M, J) \to \mathrm{Hom}_R(M', J) \to \mathrm{Hom}_R(M'', J) \\ \to \mathrm{Ext}^1_R(M, J) \to \mathrm{Ext}^1_R(M', J) \to \mathrm{Ext}^1_R(M'', J) \to \ldots \end{aligned}$$

of Algebra, Lemma 065P. Thus we see that (2) implies (1). Conversely, if J is injective then the Ext-group is zero by Homology, Lemma 0136 and Lemma 0AUL. □